

Introduction

The rapid growth of digital media and online news platforms has made information easily accessible to people across the world. Today, users can read news instantly through websites, mobile applications, and social media platforms. However, this convenience has also led to the widespread circulation of fake news, which can mislead the public, influence opinions, and create social unrest. Fake news refers to false or misleading information presented as legitimate news.

The increasing dependency on online platforms for information has made it difficult for users to distinguish between real and fake content. Misinformation spreads faster than verified news due to social sharing, sensational headlines, and the lack of proper verification mechanisms. This creates serious social, political, and economic challenges. Therefore, an automated and intelligent system is required to detect and prevent the spread of fake news.

Fake news has become a major global issue in recent years, especially with the rise of social media and digital journalism. Many individuals rely entirely on online news without verifying the source or authenticity. This project designs a system that not only detects fake news using deep learning but also provides a practical solution accessible through a web-based platform. The system ensures ease of access by offering a centralized interface where users can both read categorized news and instantly verify its authenticity without requiring technical knowledge.

To address this problem, a web-based system is developed that integrates news aggregation and fake news detection into a single platform. The system functions as a unified news aggregation and verification portal. It collects news articles from multiple trusted sources, categorizes them into different domains, displays them within a single interface, and simultaneously verifies their authenticity using an LSTM-based deep learning model. In addition, the platform allows users to manually input news content to check whether it is real or fake.

The system operates by scraping news articles from trusted online newspapers. The extracted information includes the title, content, category (such as politics, sports, technology, etc.), and source details. This ensures transparency and enables users to identify the origin of the news. The collected textual content is processed using Natural Language Processing (NLP) techniques such as tokenization, stopword removal, and text cleaning. After preprocessing, the text is converted into numerical sequences and fed into a Long Short-Term Memory (LSTM) model implemented using TensorFlow and Keras.

The LSTM model analyzes contextual and sequential patterns within the news text and predicts whether the news is real or fake. The prediction result is clearly displayed in the web portal along with the news title, category, and source name. This provides two key advantages: access to categorized news from multiple trusted sources in a single platform, and immediate verification of news authenticity.

The web application is developed using Python and Django, ensuring a user-friendly and interactive interface. Users can browse categorized news articles or manually paste content from external sources to verify authenticity. This enhances usability and makes the system practical and accessible for real-world application.

In conclusion, this project aims to reduce the impact of misinformation by integrating web scraping, natural language processing, and deep learning into a single intelligent news verification platform. The system provides a scalable and efficient solution for detecting fake news while offering a centralized portal for accessing categorized news from multiple trusted sources.